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EE-102, Section 003

Lab 3 : Combinational Logic Circuit

Experiment 3

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AIM:

Our aim in this experiment is to learn how to design the combinational circuit to simulate real-life scenario that we designed in the previous laboratory experiment with logic gates and LEDs on the breadboard. In addition, in this experiment we will learn the working principles of logic gates such as x, y, z and the details of how to integrate them into the circuit.

METHODOLOGY:

Before starting our experiment, we need to examine the details of the 4-bit binary counter (74HC/HCT163) provided to us. This stage is very important in order to successfully conclude the experiment in which we use this equipment that we do not know how to use. We use 4-bit counter to simulate the cycle of circuit and also use its counter mode while doing this. The pin configuration of counter for the work in counter mode correctly is as shown in (Figure1). Moreover, the pin configuration descriptions in (Table1) and the pin diagram on the 5-bit counter (74HC/HCT163) are as shown in (Figure2). In the experiment, we applied a 5Vpp 1 Hz square wave signal that is generated by our signal generator throughout the experiment.

MR	Pin1	High	
CP	Pin2	Signal Generator	1Hz, 5Vpp, Square Wave
D0, D1, D2, D3	Pin3,4,5,6	Ground	It does not make any difference to not connect them to ground
CEP	Pin7	High	
GND	Pin8	Ground	
PE	Pin9	High	
CET	Pin10	High	
Q0, Q1, Q2, Q3	Pin11,12,13,14	Data Outputs	They will be the inputs of the logic gates
TC	Pin15	Ground	It does not make any difference to not connect this to ground
Vcc	Pin16	High	

Table 1 : Pin configuration table

FUNCTION TABLE

OPERATING MODE	INPUTS						OUTPUTS	
	MR	CP	CEP	CET	PE	D _n	Q _n	TC
reset (clear)	L	↑	X	X	X	X	L	L
parallel load	h	↑	X	X	L	L	L	L
	h	↑	X	X	L	h	H	(1)
count	h	↑	h	h	h	X	count	(1)
hold (do nothing)	h	X	L	X	h	X	q _n	(1)
	h	X	X	L	h	X	q _n	L

Notes

- The TC output is HIGH when CET is HIGH and the counter is at terminal count (HHHH).
H = HIGH voltage level
h = HIGH voltage level one set-up time prior to the LOW-to-HIGH CP transition
L = LOW voltage level
l = LOW voltage level one set-up time prior to the LOW-to-HIGH CP transition
q = lower case letters indicate the state of the referenced output one set-up time prior to the LOW-to-HIGH CP transition
X = don't care
↑ = LOW-to-HIGH CP transition

Figure 1 : Count mode pin configuration

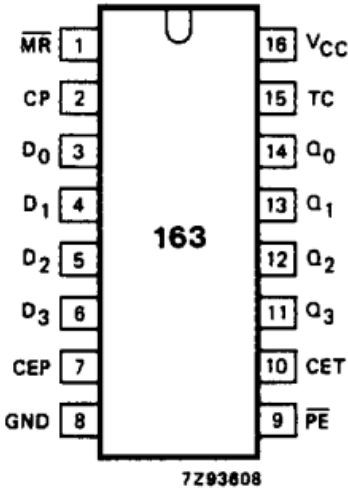


Figure 2 : Pin diagram of 74HC/HCT163

Used Logic Gates and Brief Descriptions

74HCT163 (4-Bit Binary Counter):

It operates in counter mode on a specific pin configuration (Table 1). This allows us to synchronously display the different possible output signals of the 4 inputs we have. In this experiment, we used this element as the input generator.

74LS/HC 08 (Quad 2-input AND gate):

This is a 14 pin integrated circuit containing 4 different AND gates. It contains Vcc and GND pins. In this experiment, 2 of these elements were used and the Q0 and Q1 pins of the counter were connected then outputs of the first AND gate and NOR gate were connected to second one. And the connection diagram is as in (Figure 2.1).

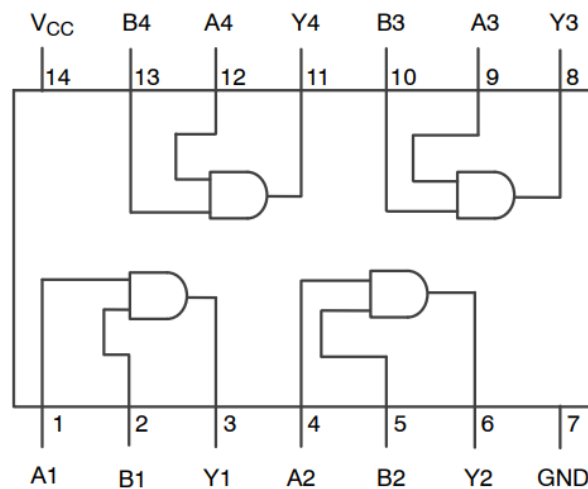


Figure 2.1 : AND gate connection diagram

74LS/HC 02 (Quad 2-input NOR gate):

This is a 14 pin integrated circuit containing 4 different NOR gates. Contains Vcc and GND pins. In this experiment, one of this element was used and the Q2 and Q3 pins of the counter were connected. And the connection diagram is as in figure 2.2.

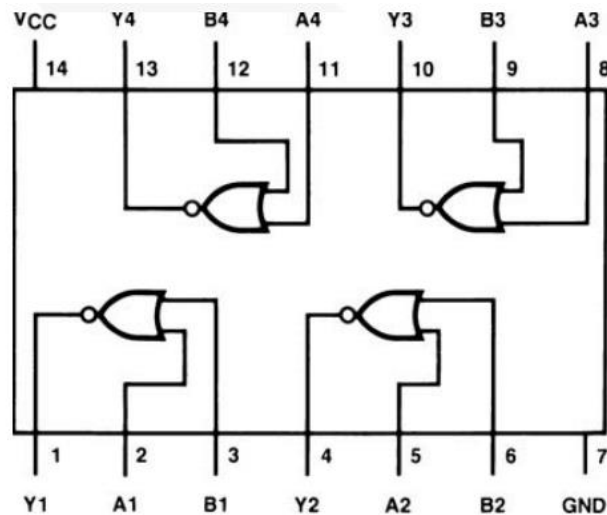


Figure 2.2 : NOR gate connection diagram

DESIGN SPECIFICATIONS:

The logic circuit we used in this experiment was built to simulate a real-life scenario. Same as last week, it aims to explain the health status of a pepper plant. Our design had 4 inputs (sunlight, water, poison, aphid) and 1 output (health). We used 2 AND Gates (74HC08) and 1 NOR gate (74 LS/HC 02) to design the logic circuit that:

$$\text{health} = (\text{sunlight AND water}) \text{ AND } (\text{poison NOR aphid})$$

The pin correspondences of these inputs and outputs on the 4-bit counter are as follows:

Sunlight	: Q0
Water	: Q1
Poison	: Q2
Aphid	: Q3
Health	: Out1

After setting up the counter inputs as desired, we started integrating the logic Gates into our breadboard. First of all, the pin configurations of the AND and NOR gates we have are as shown in (Figure 2.1) and (Figure 2.2) respectively. We installed our AND gate into the system by following the pin configurations shown in (Figure 2.1) Therefore, I connected Vcc (pin14) to 5V DC voltage and GND (pin7) to ground. Then we connected the pin named Q0 (pin 14) , which is the output of our counter, to the pin named A (pin1) of our first AND gate. Similarly, Q1 (pin13) of 4-bit counter output to the B1 (pin2) which is input our first AND gate. Accordingly, we installed the NOR gate, which we see in the connection diagram in (Figure 2.2), and connected Vcc (pin14) to receive 5V DC voltage then, we grounded GND (pin7). Furthermore, we connected the Q2 (pin 12) and Q3 (pin 11) outputs on the counter to the A1 (pin 2) and B1 (pin 3) input pins on the NOR gate, respectively. Finally, we place our 2nd AND gate on the breadboard. Likewise, we connect Vcc (pin14) to 5V DC voltage and GND (pin 7) to ground. Then, we connected the Y1 (pin 3) output on the first AND gate to the A1 (pin 1) input on the 2nd AND gate, and the Y1 (pin 1) output on the NOR gate to the B1 (pin 2) input on the 2nd AND gate. Finally, we connect a jumper cable from the Y1 (pin 3) output of the 2nd AND gate to Out1, which we call health. In order to observe how the circuit we design works and to check whether it complies with the truth table we created (Table 2) and the design scheme (Figure 4), we connected the green LEDs for each input (Q0, Q1, Q2, Q3) and the red LEDs for the output (Out1) to the breadboard along with the resistors. The final image of the circuit we created is as seen in (Figure 3) ,then we used an oscilloscope to see the output waveform.

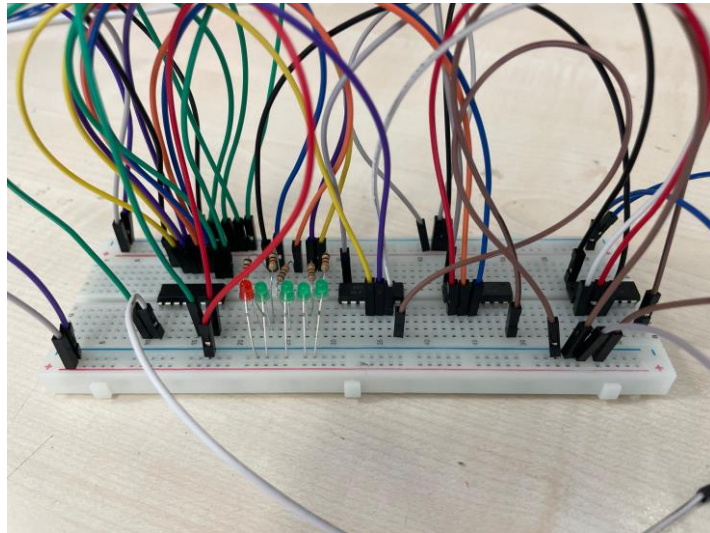


Figure 3 : Final breadboard design

And here are the truth table and design scheme for our real-life simulation circuit:

Sunlight (Q0)	Water (Q1)	Poison (Q2)	Aphid (Q3)	Health (Out1)
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Table 2 : Truth Table

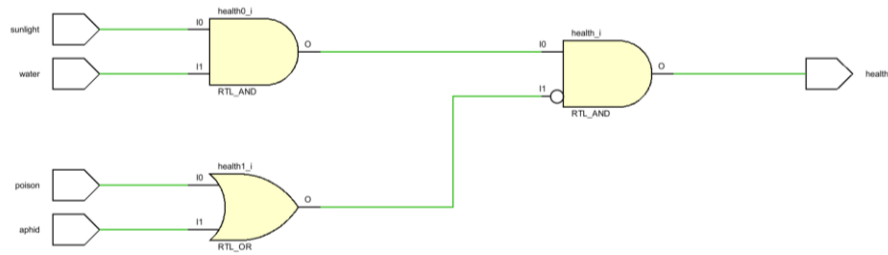


Figure 4 : Design Scheme

RESULTS:

Before starting our experiment, we created a truth table (Table 1) to observe the expected results. This table shows us that out of 16 different possibilities, only 1 of our Out1 signals will respond with a '1' signal. Then we created the circuit as stated in the design specifications. Afterwards, I set up the signal generator at 1Hz, 5Vpp and 2.5 offset, connected the positive end to the jumper cable coming from the CP (pin 2) of the counter, and the negative end to the ground, and then supplied 5V power to the breadboard. To observe the waveform, I connected my oscilloscope probe to the LED appropriately and watched the waveform. As seen in (Figure 5), the waveform we observe in the oscilloscope probe does not react in the '0' state for a long time and creates a pulse in the form of a square waveform in only 1 case, which represents the '1' state. This shows us that we have successfully built the circuit and it works as expected.

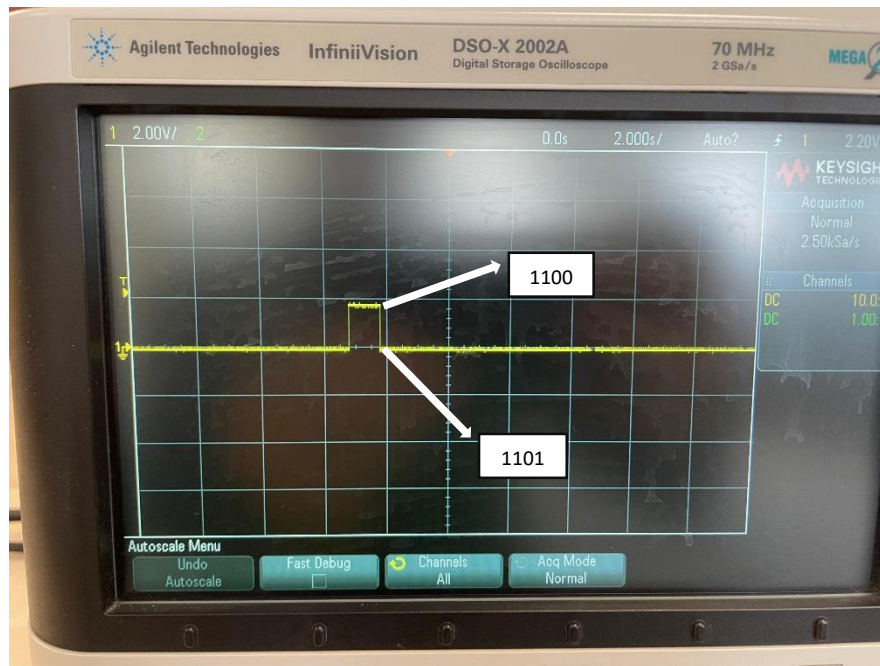


Figure 5 : Output Oscilloscope Waveform

Next, we need to test our circuit based on the truth table we created and the lighting status of the LEDs.

Combination 1 (Figure 6)

Sunlight : 1

Water : 1

Poison : 1

Aphid : 1

Expected health status : 0

Experimental health status : 0

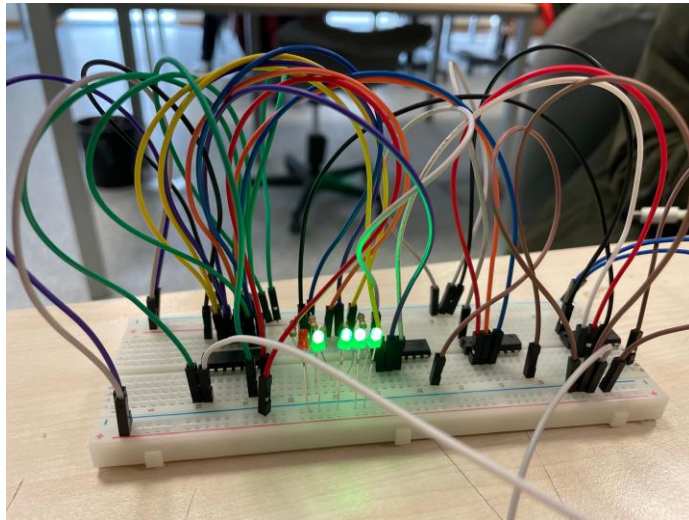


Figure 6

Combination 2 (Figure 7)

Sunlight : 0

Water : 0

Poison : 0

Aphid : 0

Expected health status : 0

Experimental health status : 0

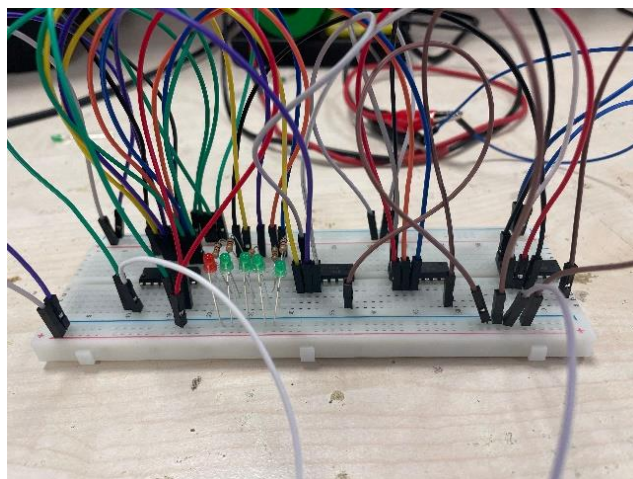


Figure 7

Combination 3 (Figure 8)

Sunlight : 1

Water : 1
Poison : 0
Aphid : 0
Expected health status : 1
Experimental health status : 1

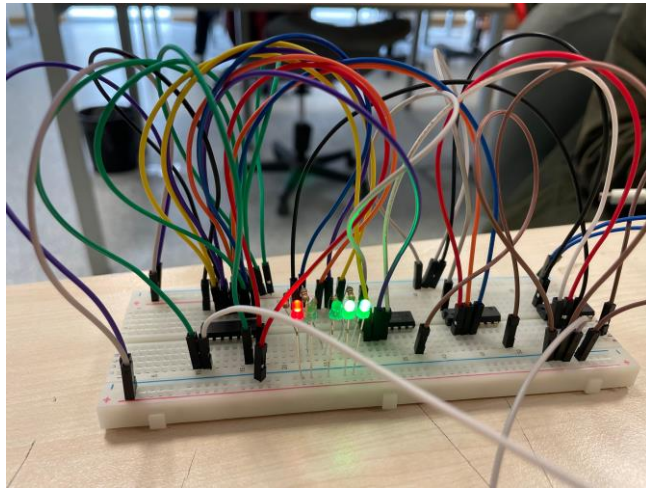


Figure 8

Combination 4 (Figure 9)

Sunlight : 1
Water : 1
Poison : 1
Aphid : 0
Expected health status : 0
Experimental health status : 0

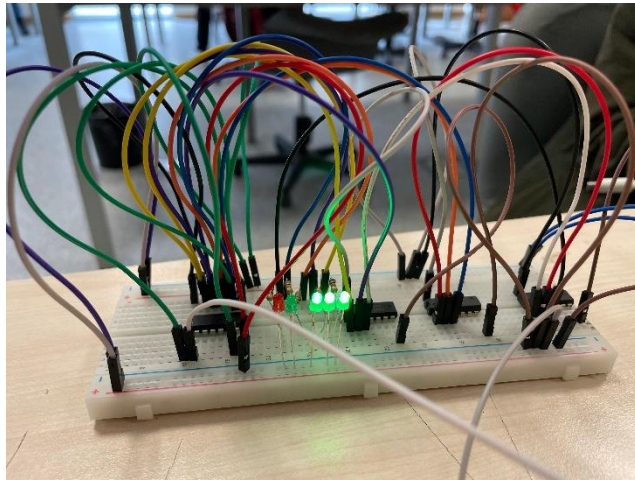


Figure 9

Combination 5 (Figure 10)

Sunlight	:	1
Water	:	0
Poison	:	0
Aphid	:	0
Expected health status	:	0
Experimental health status	:	0

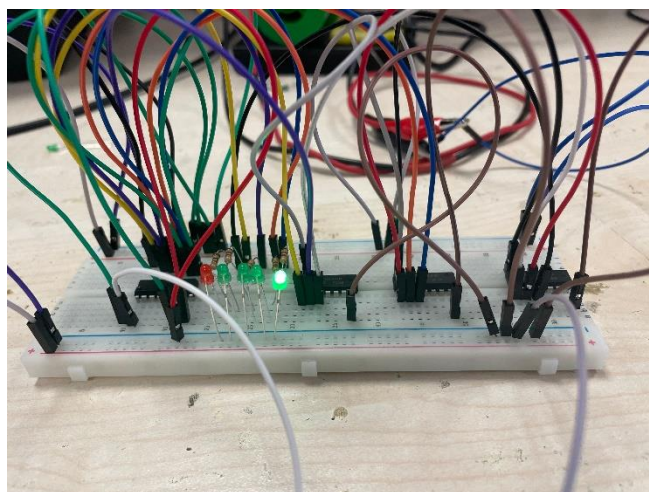


Figure 10

Combination 6 (Figure 11)

Sunlight : 1

Water : 1

Poison : 0

Aphid : 1

Expected health status : 0

Experimental health status : 0

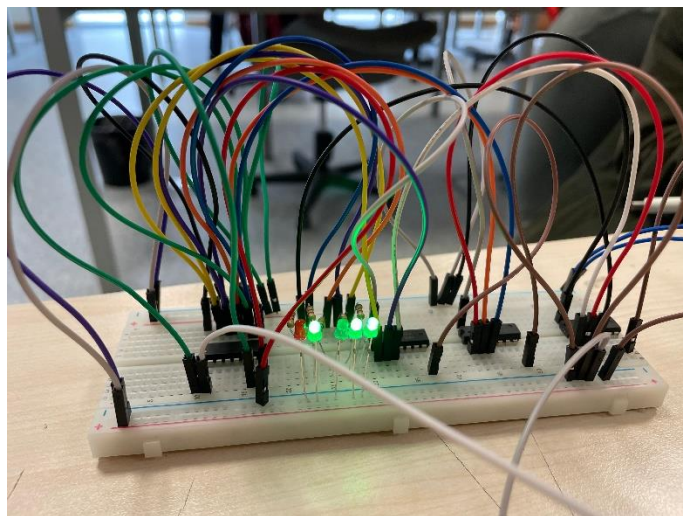


Figure 11

As can be seen, our observations were matching the expected outcomes from truth table we drew, so our experiment was successful.

CONCLUSION:

The aim of this experiment was to learn and observe how to create a combinational logic circuit by using logic gates and other circuit elements on breadboard instead of coding it with VHDL and BASYS 3. When we look at our result, our outputs showed the expected results both on the oscilloscope and in the LED output, in line with the truth table and logic design we created, and this shows that we have successfully designed this real-life simulated logic circuit. As a minor error, the wave visible on the oscilloscope appears a little later than the output LED, but this is due to the capacitors so we know that this error is minimal and insignificant. Apart from this, no other problems or errors were encountered.